INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION
IMAGE INTENSIFIER ASSEMBLY
25 MILLIMETER, MICROCHANNEL INVERTER
MX-9644/UV

This specification is approved for use by U.S. Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the 25 Millimeter, Microchannel Inverter, Image Intensifier Assembly, MX-9644/UV.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-P, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.2 Government Documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Specifications
MILITARY

MIL-PRF-49471/4	-	Battery, Rechargeable, High Performance, BA-X567/U
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Bldg #4, Section D, Philadelphia, PA 19111-5094).

2.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2.e).

American National Standards Institute

ANSI C18.1	-	Batteries and Dry Cells Specification, Specification Sheet 20-0571-1988

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

2.2.3 Reference documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.e).

DRAWINGS
USA CECOM

SM-D-850311	-	Tube Assembly, Image Intensifier, 25 MM Microchannel Inverter
SM-D-850310	-	Tube Assembly Image Intensifier
SM-D-850290	-	Power Supply, 25 Millimeter, Microchannel Inverter

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NOTE: When the Tube and Magnifier Assembly are to be purchased, reference drawing SM-D-771212-1.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Image Intensifier Assembly, 25 Millimeter, Microchannel Inverter MX-9644/UV, hereinafter designated as the assembly, shall consist of a single module containing a microchannel electron multiplier as the primary gain mechanism. The assembly shall employ an S20 photocathode with an extended red (ER) response, an aluminized phosphor screen and a power supply. The assembly shall be encapsulated and be provided with a hard surface insulating sleeve, or boot, and electrical contacts.

3.2 Construction. The assembly shall be in accordance with SM-D-850311 and all provisions of this specification.

3.2.1 Weight. The assembly weight shall not exceed 350 grams.

3.3 First article testing (FAT). When specified in the contract or purchase order (see 6.2.b), the contractor shall furnish units for FAT inspection and approval (see 4.2).

3.4 Materials. The contractor shall select the parts and materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but not mandatory. Selection criteria of the class, grade or type part shall ensure the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.5 Major Components. Major components shall be as specified herein and as shown on applicable drawings.

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3.5.1 Power supply. The power supply shall be as specified herein. Unless otherwise specified, all parts, materials and processes used in construction of the power supply shall be in accordance with SM-D-850290. The components of the power supply are identified as the regulated oscillator module and high voltage multiplier module. The oscillator/multiplier interface connection shall be in accordance with the applicable drawing. The maximum allowed dc voltage at the interface connection shall be 1,200 Vdc. Maximum alternating current (ac) voltage at the interface connection shall be 550 volts peak to peak.

3.5.1.1 Power Supply Operational characteristics. Unless otherwise specified herein, the power supply shall have the following performance characteristics at ambient conditions (see 3.5.3.12). The power supply shall meet all operational characteristics when powered by battery BA-3058/U conforming to ANSI C18.1, Specification Sheet 20-0571-1988, BA-5567/U, conforming to MIL-PRF-49471/4 or an equivalent power source

3.5.1.1.1 Voltage adjustment. The power supply shall operate in conjunction with an ON/OFF external gain adjust control (EGAC) circuit as shown in Figure 1. The maximum gain position of the EGAC shall be obtained with the switch (Sl) closed and the wiper rotated CW to position 3. The minimum gain position of the EGAC shall be obtained with the switch (Sl) closed and the wiper rotated CCW to position 1.

3.5.1.1.2 Maximum gain.

a. The microchannel plate input voltage (V_{10}) shall be not greater than 1,200 Vdc when the adjustments R_1 and R_2 are at their maximum gain settings with the following conditions applied to the power supply.
   (1) Input voltage is 3.0 Vdc.
   (2) The load at output 10 is 350 megohm.
   (3) All other outputs shall be at no load.
   (4) The EGAC is at its maximum gain position.

b. V_{10} shall be less than 700 Vdc when the adjustment R_1 varies from its maximum to minimum gain setting at the conditions as specified in the step (a) above. The adjustment R_2 shall remain at its maximum gain setting during the variation of R_1.

c. V_{10} shall be greater than 1,000 Vdc when the adjustment R_1 varies from its minimum to maximum gain setting at the following conditions applied to the power supply. The adjustment R_2 shall remain at its maximum gain setting during the variation of R_1.

   (1) Input voltage is 2.0 Vdc
   (2) The load at the output 10 is 75 megohm
   (3) All other outputs shall be at no load.
   (4) The EGAC is at its maximum gain position.

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3.5.1.1.3 Minimum gain.

a. With the EGAC at its minimum gain position and other conditions remaining as specified in 3.5.1.1.2.b, V_{10} shall be greater than or equal to 90 percent of its preset value per 3.5.1.1.2.b. When the adjustment R_2 varies from its maximum to minimum gain setting, V_{10} shall be less than 78 percent of its preset value per 3.5.1.1.2.b.

b. With the EGAC at its minimum gain position and other conditions remaining as specified in 3.5.1.1.2.c, V_{10} shall be greater than or equal to 90 percent of its preset value per 3.5.1.1.2.c. When the adjustment R_2 varies from its maximum to minimum gain setting, V_{10} shall be less than 78 percent of its preset value per 3.5.1.1.2.c.

3.5.1.1.4 Automatic brightness control (ABC) current characteristics.

3.5.1.1.4.1 ABC start current. When the power supply is operated with the EGAC at maximum gain position, the ABC start current shall be adjustable within the range as follows:

a. The ABC start current at the output 13 shall be adjustable from 200 to 450 nanoamperes (nA) by the adjustment R_3.

b. The ABC start current at the output 12 shall be adjustable from 200 to 700nA by the adjustment R_3.

c. The ABC start current in the step (a) and (b) above shall not vary more than ± 10 percent when the input voltage is changed from 3.0 to 2.0 Vdc.

3.5.1.1.4.2 ABC finish current. The ABC finish current with the EGAC at maximum and minimum gain positions shall be not greater than 50.0 nA above the ABC start current of 3.5.1.1.4.1.

3.5.1.1.5 Output regulation.

3.5.1.1.5.1 Normal mode.

a. With the EGAC at maximum gain position, and as the load current varies within the range of 0 to 40 percent of the ABC start current, V_{10} shall not vary more than ± 20.0 Vdc from its initial preset value of Table I. (The initial preset value is defined as the output voltage of V_{10} when the load current is equal to 25nA or less and the applied input voltage is 3.0 Vdc).

b. With EGAC at minimum gain position, and as the load current varies within the range of 0 to 40 percent of the ABC start current, V_{10} shall not vary more than ± 25.0 Vdc from its established voltage difference specified in 3.5.1.1.3.

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c. The output voltage requirements at outputs 5, 6, 7, 8, 9, 12 and 13 (see Figure 2) shall be within the prescribed limits of Table I. These requirements shall be met when the power supply is operated with the EGAC at maximum and at minimum gain positions.

3.5.1.1.5.2 High light resolution mode. The output voltages V_5 V_6, V_7, V_8, V_9, V_{10}, V_{12}, and V_{13} shall be as specified in Table II. These requirements shall be met when the power supply is operated with the EGAC at maximum and minimum gain positions.

3.5.1.1.5.3 Bright source protection mode.

a. With the EGAC at maximum position, V_{10} shall not exceed its preset value of 3.5.1.1.5.1.a by more than 50.0 Vdc.

b. The output voltages V_5 V_6, V_{12} and V_{13} shall be as specified in Table III. These requirements shall be met when the power supply is operated with the EGAC at maximum and minimum gain positions.

TABLE I. Output regulation, normal mode

Output Terminal	Temperature +52°C	Temperature +23°C	Temperature -54°C
5	2250 ≤ V_5 ≤ 3500 Vdc @ no load	2500 ≤ V_5 ≤ 3500 Vdc @ no load	2250 ≤ V_5 ≤ 2500 Vdc @ no load
6	1500 ≤ V_6 ≤ 2000 Vdc @ no load	1600 ≤ V_6 ≤ 1900 Vdc @ no load	1500 ≤ V_6 ≤ 2000 Vdc @ no load
7	0.51 V_6 ≤ V_7 ≤ 0.6 V_6 @ no load	0.54 V_6 ≤ V_7 ≤ 0.58 v_6 @ no load	0.51 V_6 ≤ V_7 ≤ 0.6 V_6 @ no load
8	0.8 V_5 ≤ V_8 ≤ 0.9 V_5 @ no load	0.84 V_5 ≤ V_8 ≤ 0.9 V_5 @ no load	0.8 V_5 ≤ V_8 ≤ 0.9 V_5 @ no load
9	0.58 V_6 ≤ V_9 ≤ 0.67 V_6 @ no load	0.6 V_6 ≤ V_9 ≤ 0.65 V_6 @ no load	0.58 V_6 ≤ V_9 ≤ 0.67 @ no load
10	V_{10} @ +52°C shall be within the range of +25 Vdc and -20 Vdc of the initial preset value set at +23°C. The load shall be 75% ±2% of the load at +23°C.	700 ≤ V_{10} ≤ 1000 Vdc for the loads within the range of 75 to 350 megohms.	V_{10} at -54°C shall be within the range of ±20 Vdc of the initial preset value set at +23°C. The load shall be 215% ±2% of the load at +23°C.
12	4650 ≤ V_{12} ≤ 5500 Vdc, and V_{12} at +52°C shall be within ±250 Vdc of the preset value at +23°C for a load current less than 40% of the ABC start current.	4800 ≤ V_{12} ≤ 5300 Vdc for a load current less than 40% Vdc of the ABC start current.	4950 ≤ V_{12} ≤ 5500 and V_{12} at +52°C shall be within ±250 Vdc of the preset value at +23°C f or a load current less than 40% of the ABC start current.
13	5500 ≤ V_{13} ≤ 6250 and V_{13} at +52°C shall be within ±250 Vdc of the preset value at +23°C for a load current less than 40% of the ABC start current.	5750 ≤ V_{13} ≤ 6250 Vdc for a load current less than 40% of the ABC start current.	5500 ≤ V_{13} ≤ 6250 Vdc shall be within ±250 Vdc of the preset value at +23°C for a load current less than 40% of the ABC start current.

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TABLE II. Output regulation, high light resolution mode.

Output Temperature Terminal	Temperature +52°C	Temperature +23°C	Temperature -54°C
5	$V_5 \geq 500$ Vdc for a load current greater than 205 microamperes (uA) and less than 206 uA.		
6	$V_6 \geq 400$ Vdc for a load current greater than 206 uA and less than 206 uA.		
7	$V_7 \geq 300$ Vdc at no load.		
8	$0.4\ V_5 \leq V_8 \leq 0.8\ V_5$ at no load.		
9	$0.4\ V_6 \leq V_9 \leq V_6$ at no load.		
10	$V_{10} \leq 200$ Vdc. The load shall be 75% ±2% of the load at +23°C.	$V_{10} \leq 200$ Vdc for the load between 75 to 350 megohms.	$V_{10} \leq 200$ Vdc. The load shall be 215% ±2% of the load at +23°C.
11	For a load current greater than ABC finish current, V_{12} shall be ±250 Vdc of the preset value at +23°C and also meet the requirement of $4650 \leq V_{12} \leq 5500$ Vdc.	$4800 \leq V_{12} \leq 5300$ Vdc for a load current greater than the ABC finish current.	For a load current greater than ABC finish current, V_{12} shall be ±250 Vdc of the preset value at +23°C and also meet the requirement of $4650 \leq V_{12} \leq 5500$ Vdc.
12	For a load current greater than ABC finish current, V12 shall be ±250 Vdc of the preset value at +23°C and also meet the requirement of $4650 \leq V_{13} \leq 250$ Vdc.	$5750 \leq V_{13} \leq 5300$ Vdc for a load current greater than the ABC finish current.	for a load current greater than ABC finish current, V_{12} shall be ±250 Vdc of the preset value at +23°C and also meet the requirement of $5500 \leq V_{13} \leq 6250$ Vdc.

TABLE III. Output regulation, bright source protection mode.

Output Terminal	Temperature +52°C	Temperature +23°C	Temperature -54°C
5	$0 \leq V_5 \leq 6.0$ Vdc for a load current greater than 2.6 uA and less than 6 uA.		
6	$0 \leq V_6 \leq 8.0$ Vdc for a load current greater than 206 uA and less than 8 uA.		
12	$V_{12} \leq 5500$ Vdc at no load	$V_{12} \leq 5300$ Vdc at no load	$V_{12} \leq 5500$ Vdc at no load.
13	$V_{13} \leq 6250$ Vdc at no load	$V_{13} \leq 6250$ Vdc at no load	$V_{13} \leq 6250$ Vdc at no load.

3.5.1.1.6 Input current characteristics.

3.5.1.1.6.1 Maximum peak starting (MPS) and maximum peak steady state (MPSS). The MPS input current shall not exceed 300 milliampere (mA) and the MPSS input current shall not exceed 100 mA.

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3.5.1.1.6.2 Maximum average steady state (MASS). The MASS input current shall not exceed 32mA.

3.5.1.1.7 Rise time. The rise time (see 3.5.3.15) of the power supply shall be not greater than 3 seconds, when the power supply is operated with the intensifier module from the Image Intensifier, Night Vision MX-9644/UV. The voltage overshoot of V_{10} shall not exceed 20 Vdc of its normal mode steady state value.

3.5.1.1.8 Response time (time dependent bright source). When the power supply is operating the intensifier module from Image Intensifier, Night Vision MX-9644/UV, which is subjected to a periodic pulsed input illumination in such a manner as to produce photocurrents not less than those specified in profile (a) and (b) as shown in Figures 3 and 4, the power supply shall have the following response time with the EGAC at the maximum position.

3.5.1.1.8.1 Recovery time. profile (a). The power supply shall recover to within 10 volts of its steady-state output voltage in 200 milliseconds or less after the end of the applied illumination, profile (a). Voltage overshoot of V_{10} shall be not greater than 20 Vdc of its normal mode steady state value.

3.5.1.1.8.2 Recovery time, profile (b). The power supply shall recover to within 10 volts of its steady-state output voltages in 75 milliseconds or less after the end of the applied illumination in profile (b). Voltage overshoot of V_{10} shall be not greater than 20 Vdc of its normal mode steady state value.

3.5.1.1.9 Ripple voltage.

a. The peak to peak ripple voltage at each output shall not exceed 3 percent (5 percent for V_{10}) of its steady state value when the power supply is operated in the normal mode.

b. The peak to peak ripple voltage shall not exceed 6 volts for V_5 and V_6 when the power supply is operated in the bright source protection mode.

3.5.1.1.10 Audio noise. Maximum sound pressure levels emanating from the power supply measured at a distance of 6 inches, shall not exceed the values tabulated below, against a background noise level at least 10 decibels below the sound pressure level to be measured. This requirement shall not be applicable for power supplies whose operating frequencies are greater than 22.4kHz.

Frequency		
Center (Hz)	Octave Band (Hz)	Maximum Sound Pressure Level (dB: ref 0.0002 microbars)
8000	5600-11200	17
16000	11200-22400	16

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3.5.1.1.11 Operational stability.

a. The power supply shall not exhibit arcing. There shall be no evidence of fluctuation or oscillation of the MASS input current greater than ± 0.5 mA.

b. V_{10} shall not fluctuate greater than ± 5 volts from its steady state value and shall not drift greater than ± 5 volts from its steady state value. Periodic fluctuations not greater than the ripple voltage requirements (see 3.5.1.1.9) shall be allowed only at the V_{10} oscillator frequency. All other fluctuations shall be random.

3.5.1.1.12 Operational safety.

3.5.1.1.12.1 Short circuit protection. While operating, the power supply shall not be damaged (see 3.5.3.1), when subjected to a direct short circuit of the high voltage output terminals to ground for at least one minute.

3.5.1.1.12.2 Reverse polarity. The power supply shall withstand a reverse polarity input voltage of 3.0 Vdc for at least 10 minutes. When the correct polarity of 3.0 Vdc input is applied to the power supply, the power supply shall conform to the requirement of 3.5.1.1.6.2.

3.5.1.1.13 Electromagnetic interference. The power supply shall comply with the following Class A3 requirements of MIL-STD-461: RE01, RE02, RS01, RS02 and RS03 except that the distance for measuring RE01 shall be 42 centimeter (cm). The test shall be performed on a simulated intensifier module and inside the tube housing operated at ground potential. A change in MASS input current, which shall be measured during susceptibility testing, greater than 1.0 mA during the susceptibility tests RS01, RS02 and RS03 shall constitute the degradation of power supply performance. The frequency to be tested under RE02 shall be 14khz - 10 Ghz. The limitations on RE02 are as follows:

20.5 - 35.5 Khz - 35db
35.5 - 88.0 Khz - 45db
88.0 - 130.0 Khz - 25db
130.0 - 340.0 Khz - 20db

The Radiated Susceptibility frequencies under RS03 are as follows:

a. 10 KHz - 2MHz
b. 2 MHz - 30MHz
c. 30 MHz - 10GHz
d. 10 GHz - 40GHz

3.5.1.1.14 Reliability. The power supply shall have a minimum O_1, mean time between failure (MTBF) of 9000 hours, tested in accordance with approved test procedures.

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3.5.2 MX-9644/UV Image intensifier assembly operational and environmental requirements. The assembly shall operate in conjunction with a nominal 50 thousand ohm gain control switch, and a 470 thousand ohm resistor in accordance with Figure 1. The minimum manual gain setting shall be obtained with the switch (Sl) closed, and the wiper rotated CCW to position 1. The maximum manual gain setting shall be obtained with the switch (Sl) closed and the wiper rotated CW to position 3. The voltage source for operating the assembly shall be from a BA-3058/U battery conforming to ANSI C18.1, Specification Sheet 20-0571-1988, BA-5567/U battery conforming to MIL-PRF-49471/4 or a power source with an input of 2.0 to 3.0 volts direct current (dc).

3.5.2.1 Photocathode sensitivity. The photocathode sensitivity shall be not less than the following values for the conditions specified.

a. Luminous sensitivity of 260 microamperes (uA) per lumen for 2856 Kelvin (K) radiation.
b. Radiant sensitivity of 0.015 ampere per watt at 0.80 micrometer.
c. Radiant sensitivity of 0.010 ampere per watt at 0.85 micrometer.

3.5.2.2 Burn-in, environmental stress screening (ESS). The assembly shall undergo cyclic operation (burn-in) of not less than 120 continuous 12 minute cycles. Each cycle of operation shall have a profile as specified in Figure 2. At the end of this cyclic operation there shall not be evidence of damage (see 3.5.3.1). Burn-in shall be performed with operating voltage applied. The photocathode shall be illuminated at least once during each 12 minute cycle with 1 to 5 X 10^{-3} foot-candles for not less than 3 seconds and 5 to 10 foot-candles for not less than 5 seconds.

3.5.2.3 Vibration. With the operating potential applied, the assembly shall not be damaged, (see 3.5.3.1) after being rigidly mounted, singly or in groups, and subjected to a 2.5 g's (see 3.5.3.3) peak sine wave vibration over a frequency range of 10 hertz (Hz) through 3500 Hz and return to 10 Hz. in a logarithmic sweep within 30 minutes along both the parallel and perpendicular axes of the assembly. The sweep shall be performed 2 times in each axis.

3.5.2.4 Mechanical shock.

3.5.2.4.1 Mechanical shock I. With the operating potential applied, the assembly shall not be damaged (see 3.5.3.1) when subjected to 6 shock impacts applied parallel to the optical axes and 6 shock impacts applied perpendicular to the optical axis of the assembly. The shock impacts shall be nominal half sine wave pulse having a minimum of 500 g's (see 3.5.3.3) peak value and a duration of 0.3 milliseconds, ± 0.05 milliseconds measured between the 10 percent values of the peak amplitude.

3.5.2.4.2 Mechanical shock II. The assembly shall not be damaged (see 3.5.3.1) when subjected to 6 mechanical shock pulses along the perpendicular axis and 6 mechanical shock pulses along the parallel axis at an amplitude of 140 g's, ± 10 g's (see 3.5.3.3) half-sine wave peak for 9 milliseconds, ± 10 percent duration.

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3.5.2.5 Environmental temperature. The assembly shall not be damaged (see 3.5.3.1) by the storage, operation, or thermal shock temperature profile specified and shall meet the requirements at the temperature specified. Test profile shall be as follows:

a. High temperature storage: +68°C for 1 hour

b. Temperature/Humidity: +52°C, 95% relative humidity, 4 hours
(This portion of the test may be run separately.)

c. High Operating temperature: +52°C, <25% relative humidity, 1 hour
(1)Input current - Shall not exceed 35 milliamperes.

(2) Equivalent background input (EBI) - Not greater than 6 x 10^{-10} lumens/cm^2.

(3) Gain @ 1 to 5 x $l0^{-6}$ foot-candle (fc) - Not less than 20 thousand and not greater than 80 thousand.

(4) Screen brightness @ 5 x 10^{-4} to 1 to 10^{-3} fc - Not less than 2 footlamberts and not greater than 10 footlamberts.

(5) Screen brightness @ .01 to .03 fc - Not less than 2 footlamberts and not greater than 10 footlamberts.

(6) Operational fluctuation @ .05 to .1 fc shall be not greater than plus or minus 10 percent of steady state value for not less than 30 seconds.

(7) Rise time - Not greater than 5 seconds, overshoot less than 40 percent of steady state output brightness.

d. Temperature shock: Shock from +52°C to +25°C in less than 5 minutes.

e. Low temperature storage: Reduce temperature to -62°C, hold 1 hour

f. Low operating temperature: -54°C, 1 hour
(1) Input current - Not greater than 25 milliamperes.

(2) EBI - Not greater than 2 x 10^{-11} lumens/cm^2.

(3) Gain @ 1 to 5 x $l0^{-6}$ fc - Not less than 20 thousand and not greater than 80 thousand.

(4) Screen brightness @ 5 x 10^{-4} to 1 x 10^{-3} fc - Not less than 2 footlamberts and not greater than 10 footlamberts.

(5) Screen brightness @ .01 to .03 fc - Not less than 2 footlamberts and not greater than 10 footlamberts.

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(6) Operational fluctuation @ .05 to .1 fc shall be not greater than plus or minus 10 percent of steady state value for not less than 30 seconds.

(7) Rise time - not greater than 5 seconds, overshoot less than 40 percent of steady state output brightness.

(8) The variation between room temperature gain tests prior to high temperature and after cold temperature shall not be greater than plus or minus 15 percent. The room temperature EBI shall be less than or equal to 2 x 10^{-11} lumens/cm^2 after high and low temperature test.

g. Temperature shock: Shock from -54°C to +25°C within 5 minutes.

h. Stabilization temperature: +25°C, 1 to 16 hours

i. Battery Operation Temperature: -17.8°C, 30 minutes minimum

(1) Gain @ 1 to 5 x $l0^{-6}$ fc - Not less than 30 thousand and not greater than 70 thousand.

(2) Rise time - Not greater than 5 seconds.

3.5.2.6 Equivalent background input. The equivalent background input at room temperature shall not exceed 2 x 10^{-11} lumens/cm^2.

3.5.2.7 Luminance gain. The assembly shall have a room temperature (see 3.5.3.12) luminance gain and high light saturation characteristic as specified in Table IV. The input current shall not exceed 32 milliamperes (mA) at the light levels specified.

TABLE IV. Saturation requirements for luminance gain.

Input Manual Gain setting	Nominal Input Light Level (fc)	Minimum Allowable Gain	Maximum Allowable Gain	Minimum Allowable Output Brightness (ft-l)	Maximum Allowable Output Brightness (ft-l)	Max. Input Current (mA)
Maximum	5 X 10^{-6}	40,000	70,000	--	--	32
Maximum	5 X 10^{-4}	--	--	2	10	32
Maximum	5 X 10^{-2}	--	--	2	10	NA
Maximum	10.0	0	10^{-3}	--	--	NA
Maximum	5 X 10^{-5}	2,000	10,000	--	--	32
Maximum	5 X 10^{-3}	--	--	2	10	32

3.5.2.8 Time dependent bright source. When the assembly is subjected to a periodic pulsed input illumination in such a manner as to give rise to photo-currents in accordance with profiles (a) and (b) as shown in Figures 3 and 4, the assembly shall have response at maximum manual gain setting as set forth below.

3.5.2.8.1 Recovery time, profile (a) Figure 4. The assembly shall recover to within ± 50 percent of the ambient useful steady state output brightness (see 3.5.3.10) in 400 milliseconds or less after the end of the applied illumination, profile (a). Recovery time to the ambient useful steady state output brightness shall be not greater than 800 milliseconds.

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3.5.2.8.2 Recovery time, profile (b), Figure 5. The assembly shall recover to within ± 50 percent of the ambient useful steady state brightness in 300 milliseconds or less after the end of the applied illumination profile (b). Recovery time to the ambient useful steady state output brightness shall be not greater than 600 milliseconds.

3.5.2.8.3 Resolution, profile (b) Figure 5. With a steady state illumination input of 1 to 5 x 10^{-5} foot-candle and after 3 cycles of profile (b), the assembly shall have a minimum resolution of 16 line pairs per millimeter (lp/mm) during continuous application of profile (b).

3.5.2.9 Bright source protection. The assembly shall not be damaged (see 3.5.3.1) when the input illumination is 50 millilumens concentrated on a 1.0 square millimeter area of the photocathode during a time interval of 20 seconds, ± 1 second.

3.5.2.10 Signal-to-noise ratio. The signal-to-noise ratio (S:N) of the assembly shall be not less than 2.8 to 1.

3.5.2.11 Fixed pattern noise. Fixed pattern noise shall be as follows:

3.5.2.11.1 Multi-to-multi-pattern variation. Deviation from the mean output brightness shall not exceed ±10 percent.

3.5.2.11.2 Multi-boundary pattern noise. When sampled by a 25 micrometer photometering aperture the deviation from the mean output brightness over three boundaries shall be within the limits of plus 8 or minus 15 percent.

3.5.2.12 Output brightness uniformity. The output brightness uniformity of the assembly shall not exceed a ratio of 2 to 1 over the full 25 millimeter useful cathode diameter.

3.5.2.13 Image alignment. The center of an image produced on the screen of the assembly by focusing a test reticle on the photocathode concentric with the optical axis shall fall within a 0.040 inch diameter circle on the image screen concentric with the optical axis.

3.5.2.14 Image shift. The center of an image produced on the screen of the assembly by focusing a test reticle on the photocathode concentric with the optical axis shall shift not more than 0.005 inch from the optical axis.

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3.5.2.15 Resolution.

3.5.2.15.1 Center resolution. The center resolution of the intensifier shall be not less than 32.0 lp/mm with an input radiation adjusted for best contrast.

3.5.2.15.2 High light level center resolution. With the entire photocathode of the assembly illuminated with not less than 0.8 nor greater than 5 foot-candles the minimum high light level center resolution shall be 16 lp/mm.

3.5.2.15.3 Peripheral resolution. The peripheral resolution of the assembly shall be not less than 28 lp/mm at four points, l0mm, ± 0.5mm from the center of the photocathode with an input radiation adjusted for best contrast.

3.5.2.16 Modulation transfer function. With an input light level of less than or equal to 2 x 10^{-3} foot-candle, the modulation transfer function (MTF) (see 3.5.3.6) at the center of the image screen shall be not less than:

a. 90 percent MTF at 2.5 lp/mm.
b. 60 percent MTF at 7.5 lp/mm.
c. 25 percent MTF at 15 lp/mm.

3.5.2.17 Useful cathode diameter. The useful cathode diameter shall be not less than 25 millimeters.

3.5.2.18 Magnification. The magnification shall be not less than 0.96, nor greater than 1.04 over the useful cathode area. The magnification shall not vary more than 5 percent over the useful cathode area.

3.5.2.19 Shear distortion. Shear distortion (see 3.5.3.9) of the assembly shall not be greater than 50 micrometers at any point within 0.8 inch diameter quality circle concentric with optical axis of the assembly.

3.5.2.20 Electrical stability. The assembly shall show no evidence of damage when the operating assembly is probed with a 1/4 inch diameter ground probe. The input current shall not deviate more than 0.5 milliamperes.

3.5.2.21 Phosphor decay. With an input illumination of 1 to 5 x 10^{-5} foot-candle, the decay due to the output phosphor screen response shall be within the limits specified in Figure 5 at 10 milliseconds, 100 milliseconds, 500 milliseconds, and 1 second from the start of decay.

3.5.2.22 Phosphor spectral response. The peak relative spectral response of the output screen shall occur at a wavelength between 5100 and 5600 angstroms. The bandwidth shall be less than 2000 angstroms measured at the 10 percent points of the spectral response curve. The relative response at 6500 angstroms shall be not greater than 10 percent of the nominal peak response.

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3.5.2.23 Photocathode and screen quality. When the screen of the assembly is viewed with a l0X or greater magnifier and with the radiation level on the photocathode adjusted to obtain best spot contrast, the bright spots, opaque spots, or dark spots which exceed a contrast of 30 percent of their surrounding area shall not exceed the size and quantities specified in Table V. The size of non-circular spots shall be determined on the basis of equal area to circular spots. When the distance between two spots is less than the maximum dimension of either spot, the two spots shall be considered as one spot with size equal to the sum of the maximum dimensions of the two spots plus the amount of separation between them. Graininess shall not be discernible anywhere over the useful diameter of the screen. There shall be no field emissions (see 3.5.3.5) except as allowed in Table V or ion spots (see 3.5.3.4) discernible on the output phosphor screen when viewed with a 10X magnifier after the input photocathode of the assembly has been illuminated with an initial level of not less than 5×10^{-3} foot-candles and then reduced to 2×10^{-7} foot-candles.

TABLE V. ASSEMBLY SPOTS.

Size of Spots in Inches	Accumulative Number of spots within 0.30 Inch Diameter Circle	Accumulative Number of spots within area bounded by two circles 0.30 and 0.80 inches in diameter	Accumulative number of spots within area bounded by two circles 0.80 and 0.98 inches in diameter
Greater than 0.015	0	0	0
0.012 - 0.015 in.	0	1	2
0.009 - 0.119 in.	0	3	6
0.006 - 0.0089 in.	1	6	10
0.003 - 0.0059 in.	1	15	20
*0 - 0.0009 in.	1	3	5
*0.001 - 0.003 in.	1	3	4
*> 0.003 in.	0	0	0

* Bright spots only
1. The 0.30, 0.80, 0.98 inch circles on the image tube shall be concentric with the optical axis of the assembly.

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3.5.2.24 Reliability. The assembly shall have a minimum mean-time-to failure (MTTF) of 2500 hours when operated under the reliability test conditions specified herein. Once every 200 ±50 hours of operation, each assembly on test shall be subjected to all of the tests of Table VI. Reliability Test assemblies shall be removed from test conditions for not more than 15 hours for measurement of assembly parameters. For testing in accordance with Table VI, the acceptable minimum limit for EBI shall be 5 X 10^{-11} lumens/cm^2. Luminance gain shall be measured at 5 X 10^{-6} fc only. Acceptable limit for luminance gain shall be 20,000 at this light level. The accept/reject determination shall be made on either the accumulation of 2500 hours operating time on the sample assemblies or the occurrence of four (4) failures for FAT and two (2) failures for production testing.

TABLE VI Performance verification tests.

Test	Requirement Paragraph
Equivalent background input	3.5.3.6
Luminance gain	3.5.3.7
Signal-to-noise ratio	3.5.3.10
Damage	3.5.4.1

3.5.2.25 Identification markings. Each assembly shall be identified in accordance with drawing SM-D-850310. The markings shall include a coded acceptance date. The first two numbers of the code shall be two digits indicating the calendar week of the year (01 through 52). Reading from left to right, or, top to bottom, the code number shall indicate the year and week of acceptance in that order.

3.5.2.26 Workmanship. Standards of workmanship shall be such that the assembly will meet all requirement of this specification and any referenced specifications or drawings.

3.5.3 Technical interpretations. The following technical interpretations are, when referenced in sections 3 or 4, mandatory for this specification.

3.5.3.1 Damage. Damage is defined as electrical failure or malfunctioning including arcing, corona, flashing, bright spots more than allowed in 3.5.2.23, flickering, blinking, field emissions, ion spots, cracks, scratches, breakage, deformation, corrosion, or deterioration of any part or finish, (no more than 2 flashes in each axis) or missing or loose components.

3.5.3.2 Optical length. The optical length of the assembly is defined as the distance from the input image plane to the output image plane.

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3.5.3.3 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

3.5.3.4 Ion spot. A high concentration of ion scintillations generally in the center of the field.

3.5.3.5 Discernible field emission. Discernible field emission is extraneous emission which appears as bright spots or pattern that may flicker or appear intermittently on the image screen in one general position. Field emission is voltage dependent and is best observed with no radiation incident on the photocathode.

3.5.3.6 Modulation transfer. Modulation transfer is a measure of the degradation of a signal image as it appears at the output screen of the assembly as correlated to the input signal pattern which is normalized to 100 percent contrast at a spatial frequency equal to or less than 0.2 line pair per millimeter.

3.5.3.7 Limiting resolution. Limiting resolution is defined as the smallest resolution pattern which the observer can see and distinguish between the black lines and the clear area between the black lines. The observer must be able to determine the number of line pairs in both the vertical and horizontal test pattern.

3.5.3.8 Ion scintillations. Bright spots which occur on the image screen randomly in space and time.

3.5.3.9 Shear distortion. Shear distortion is defined as the displacement or rotation of a small line segment with respect to a perfect straight line image.

3.5.3.10 Output brightness. All output light levels refer to those levels at the output of the last fiber optic.

3.5.3.11 Optical axis. The optical axis of the assembly is defined as the mean center line of that cylindrical portion of the assembly used to align the assembly in the system housing.

3.5.3.12 Room (ambient) temperature. Room temperature is defined as +23°C, ± 10°C.

3.5.3.13 Environmental temperature. Environmental chamber temperature shall be controlled to within ±5°C of the specified temperature unless otherwise specified.

3.5.3.14 Rise time, Assembly. The rise time is the time required for the assembly to achieve 90 percent of its steady state output brightness after the voltage source is applied to the assembly.

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3.5.3.15 Rise Time, Power Supply. The time required for V_{10} to settle within 10 volts of its steady state value after the input voltage has been applied.

3.5.3.16 Ambient useful steady state output brightness. The ambient useful steady state output brightness of the assembly is defined as the steady state output brightness with the operating potentials applied, the manual gain control set at maximum, and any specified steady state input illumination applied.

4. VERIFICATION

4.1 Classification of inspection. Inspections shall be classified as follows:

a. First Article Testing (4.2).
b. Conformance inspection (4.3).

4.2 First Article Testing (FAT). Unless otherwise specified in the contract or purchase order (see 6.2.f), the FAT inspection shall be performed by the contractor. The quantity of first article samples shall be as specified in the contract or purchase order (see 6.2.b) and shall be selected from the first production lot. First article test of the power supply shall be as proposed by the contractor and approved.

4.2.1 First article testing. Each assembly in the first production lot shall be subjected to the performance tests of Table VII, which may be performed in any order. After completion of this testing, assemblies selected as samples for FAT testing shall be subjected to the tests of Table VII-A, which may be performed in any order. Failure of any test shall be cause for rejection of that assembly and may be cause for rejection of the FAT.

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TABLE VII. Unit First Article Testing.

Test	Requirement Paragraph
Workmanship	3.5.2.26
Identification marking	3.5.2.25
Photocathode sensitivity	3.5.2.1
Burn-in (ESS)	3.5.2.2
Equivalent background input	3.5.2.6
Luminance gain	3.5.2.7
Time dependent bright source	3.5.2.8
Bright source protection	3.5.2.9
Signal-to-noise ratio	3.5.2.10
Fixed pattern noise	3.5.2.11
Output brightness uniformity	3.5.2.12
Image alignment	3.5.2.13
Image shift	3.5.2.14
Center Resolution	3.5.2.15.1
High light level resolution	3.5.2.15.2
MTF	3.5.2.16
Useful cathode diameter	3.5.2.17
Magnification	3.5.2.18
Sheer distortion	3.5.2.19
Electrical stability	3.5.2.20
Photocathode and screen quality	3.5.2.23

TABLE VII-A Sample First Article Testing.

Test	Requirement Paragraph
Weight	3.2.1
Phosphor decay	3.5.2.21
Phosphor spectral response	3.5.2.22
Vibration	3.5.2.3
Mechanical shock I	3.5.2.4.1
Mechanical shock II	3.5.2.4.2
Environmental temperature	3.5.2.5
Reliability	3.5.2.24
Electromagnetic interference	3.5.1.1.13

4.2.2 Disposition of FAT samples. Disposition of FAT samples shall be as specified in the contract or purchase order (see 6.2.b).

4.3 Conformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.g). Unless otherwise specified, all tests shall be performed at ambient temperature.

4.4 Inspection lot. For purposes of inspection, a lot shall be defined by the contractor.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2.i). When actual packaging of materials is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES: (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use. The image intensifier assembly covered by this specification is intended for use in an electro-optical viewing device to intensify low light level such that a visible image is presented for viewing and sighting purposes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Quantity and schedule for FAT testing for all line items and disposition of FAT samples.
c. Necessary actions by the contractor in the event of a lot failure.
d. Sampling plan for lot testing.
e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.2.3)
f. Statement specifying that FAT testing is, or is not, required.
g. Statement identifying that the contractor must submit his proposal for conformance inspection to include the specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.
h. Statement identifying that the contractor must submit his proposal for Environmental Stress Screening (ESS).
i. Packaging requirements.

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6.3 Definitions. See 3.5.3.

6.4 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

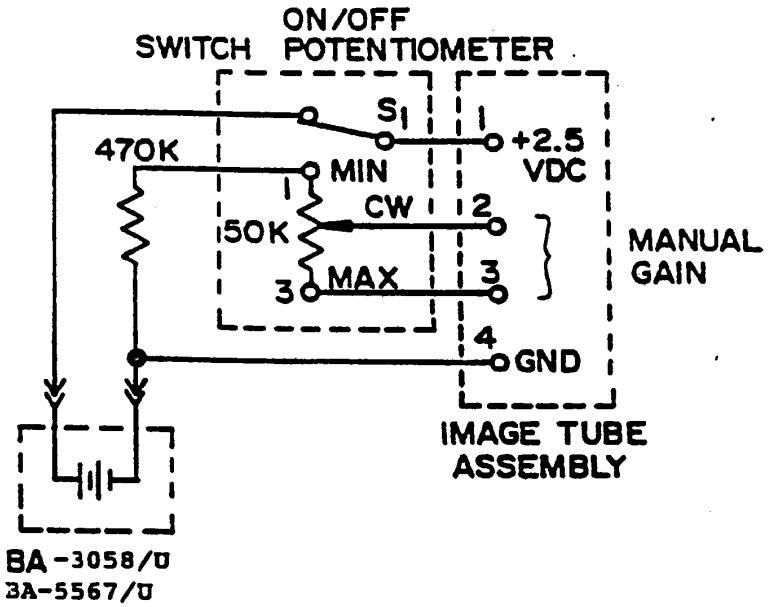
6.5 Subject term keyword listing.
Night Vision Assembly
Tube Assembly

CUSTODIAN:
Army - CR
Navy – EC

PREPARING ACTIVITY:
Army - CR

Project No. 5855-0164

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NOTES:

1. ON/OFF SWITCH POTENTIOMETER
   A. MANUFACTURER: ALLEN AND BRADLEY, PART NO. GS4T032F503 UA, OR EQUAL
   B. POWER RATING: 0.5 WATT AT 70°C AMBIENT.
   C. RESISTANCE: 50,000 OHMS ± 20 PERCENT.
   D. CURRENT RATING: 1.5 AMPERES FOR 28 VDC.

2. RESISTOR
   A. RESISTANCE: 470,000 OHMS ± 10 PERCENT.
   B. POWER: 1/4 WATT.
   C. TYPE: CARBON, FIXED COMPOSITION

FIGURE 1. Manual brightness gain control wiring diagram.

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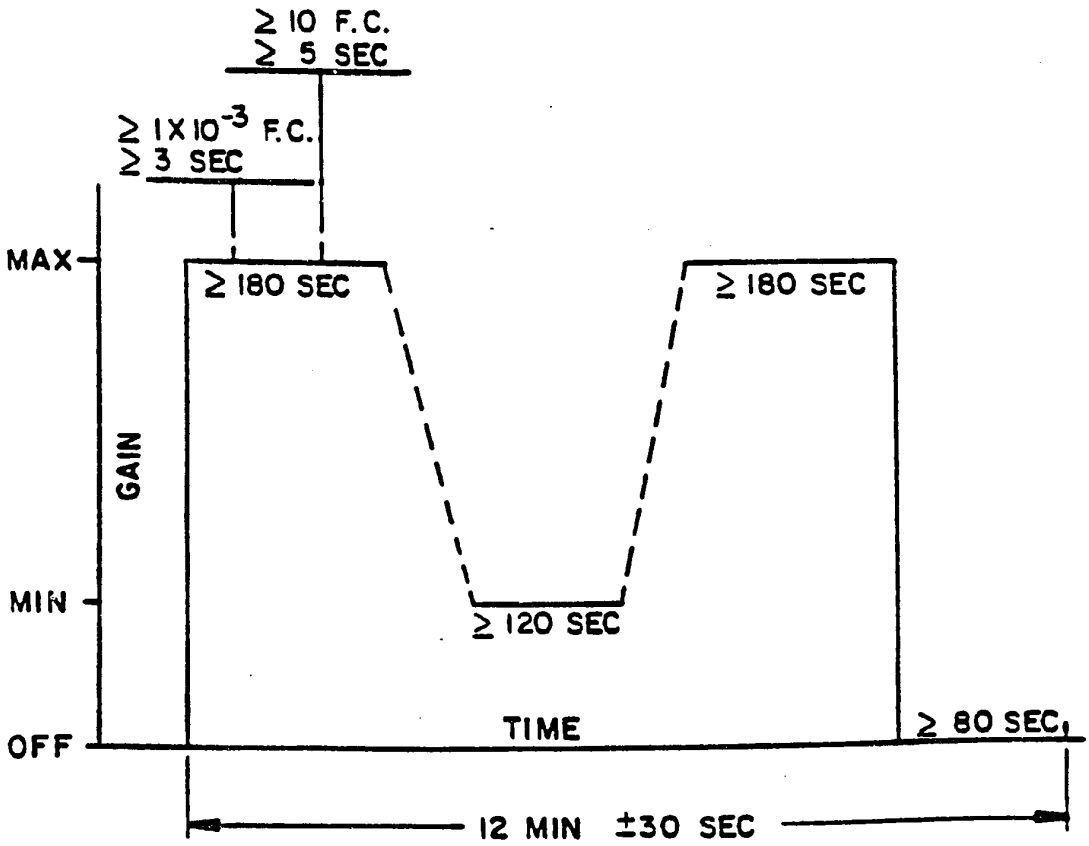


FIGURE 2. Cyclic operation, (Burn-in) Profile.

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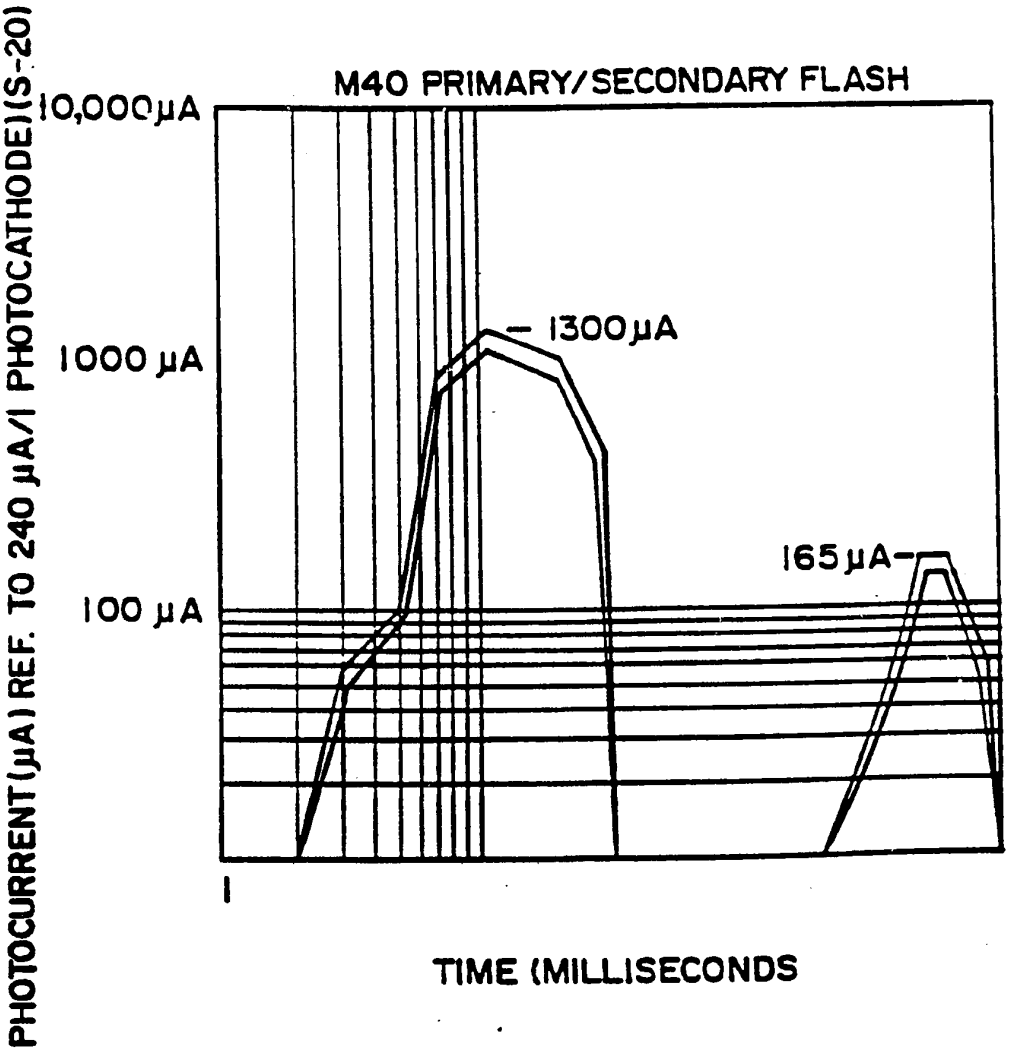


FIGURE 3. Profile (a).

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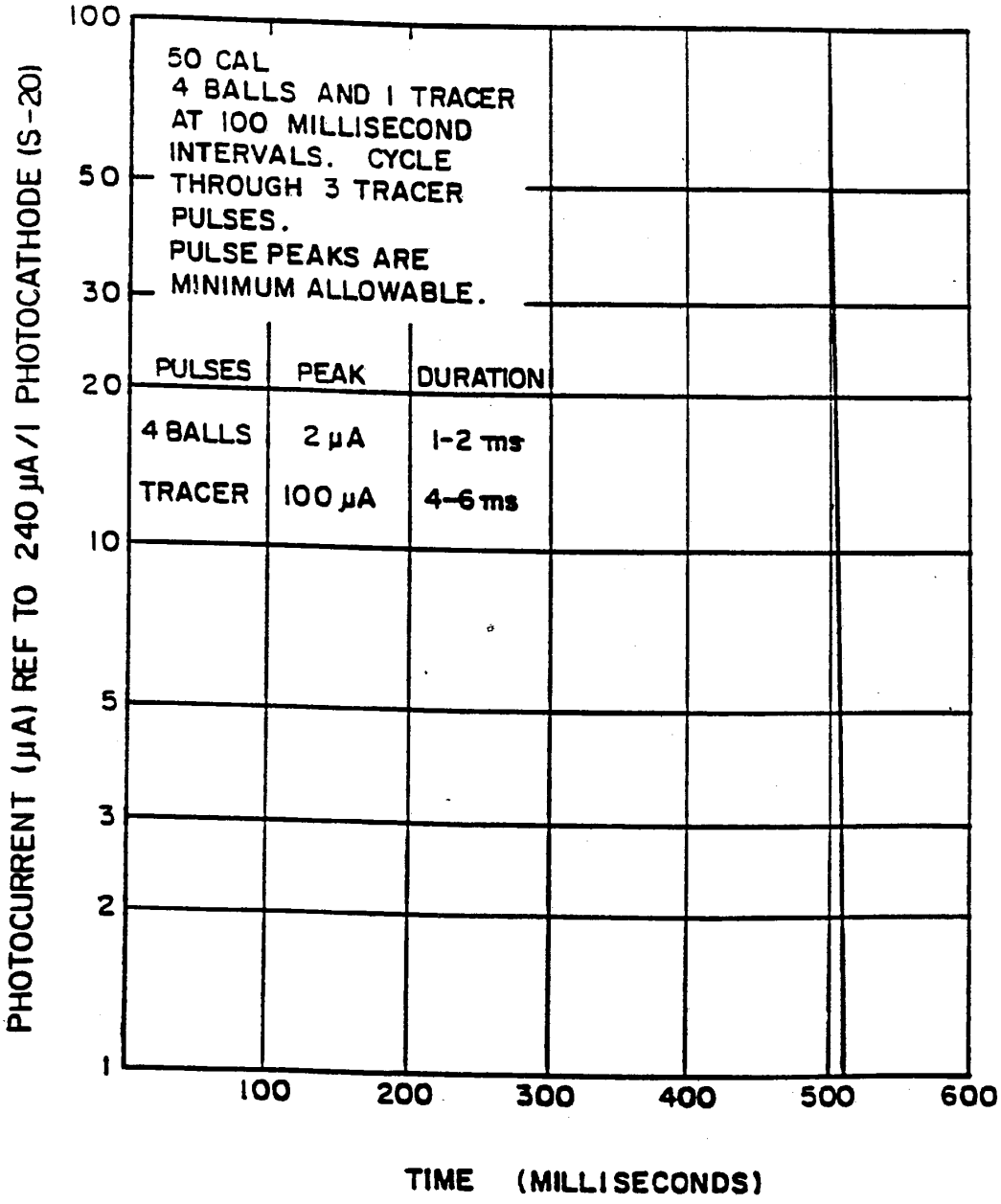


FIGURE 4. Profile (b).

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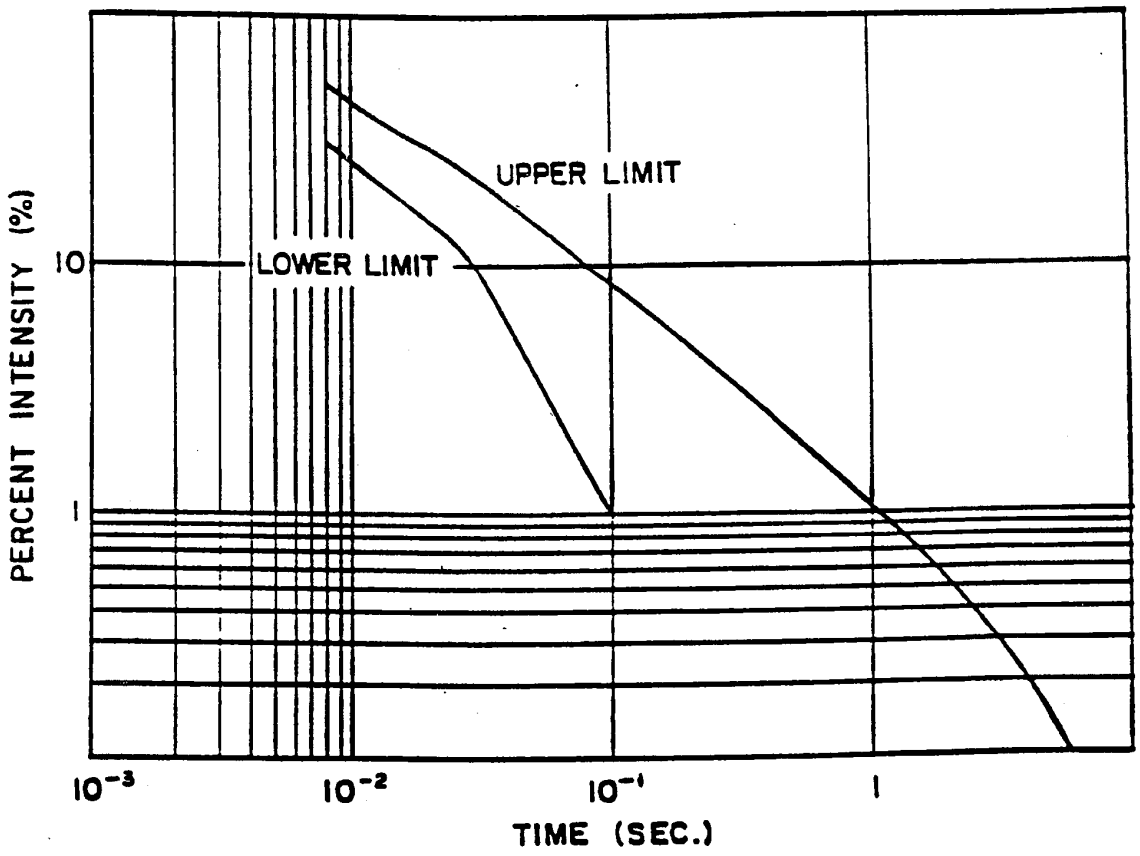


FIGURE 5. Phosphor decay spec. curves.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49040F	**2. DOCUMENT DATE (YYMMDD)** 990304

3. DOCUMENT TITLE
Image Intensifier Assembly 25 Millimeter, Microchannel Inverter MX-9444/UV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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